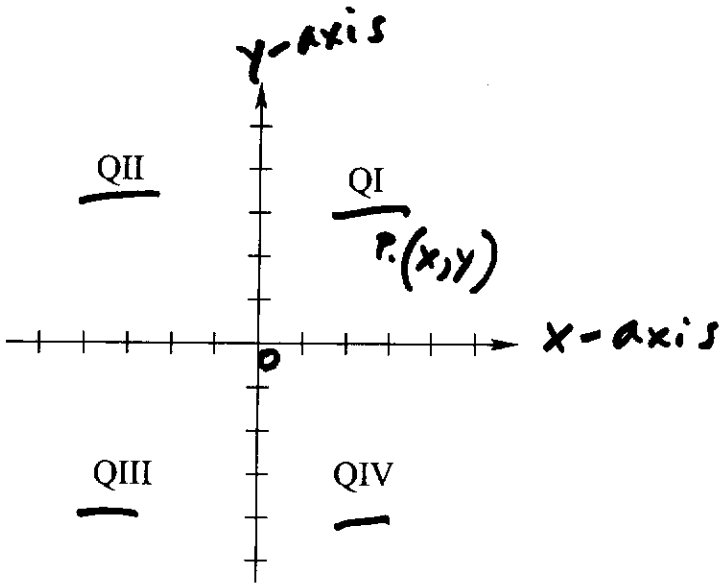
Frank 4th CARAUD

Lecture 7, Part I: Section 1.1
Rectangular Coordinates

<u>**Def.**</u> The **rectangular** or **Cartesian coordinate system** is formed by **coordinate axes:** horizontal x-axis and vertical y-axis.

The **origin** is the point of intersection of coordinate axes. the axes divide the **coordinate plane** or xy-**plane** into four **quadrants**.



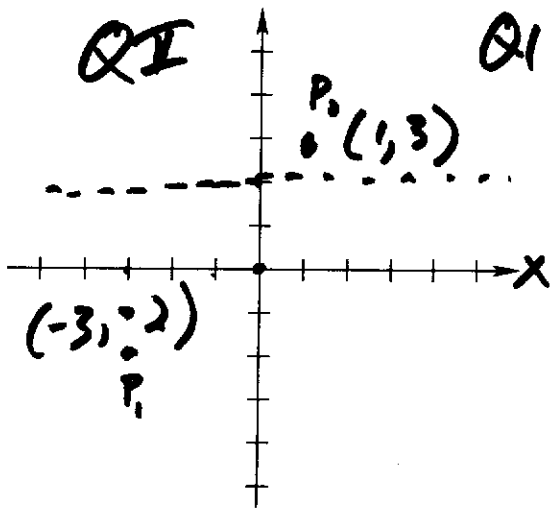
Each point P in the xy-plane corresponds to an **ordered pair** (x, y) of real numbers called the **coordinates** of P.

To **plot** $P = (x, y)$:

x-coordinate: **directed distance** of the point from the y-axis

y-coordinate: **directed distance** of the point from the x-axis

<u>**ex.**</u> Plot $P_1 = (-3, -2)$ and $P_2 = (1, 3)$.

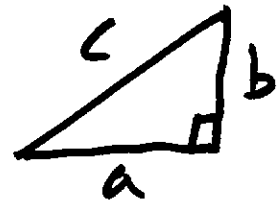


<u>**ex.**</u> Locate all points which are 2 units above the x-axis. In which quadrant(s), if any, do they lie?

$(x, 2)$ QI, QII

<u>**ex.**</u> Locate all points with x-coordinate 0.

$(0, y)$ is y-axis



Pythagorean Theorem

For a right triangle with hypotenuse of length c and sides of lengths a and b, we have $a^2 + b^2 = c^2$.

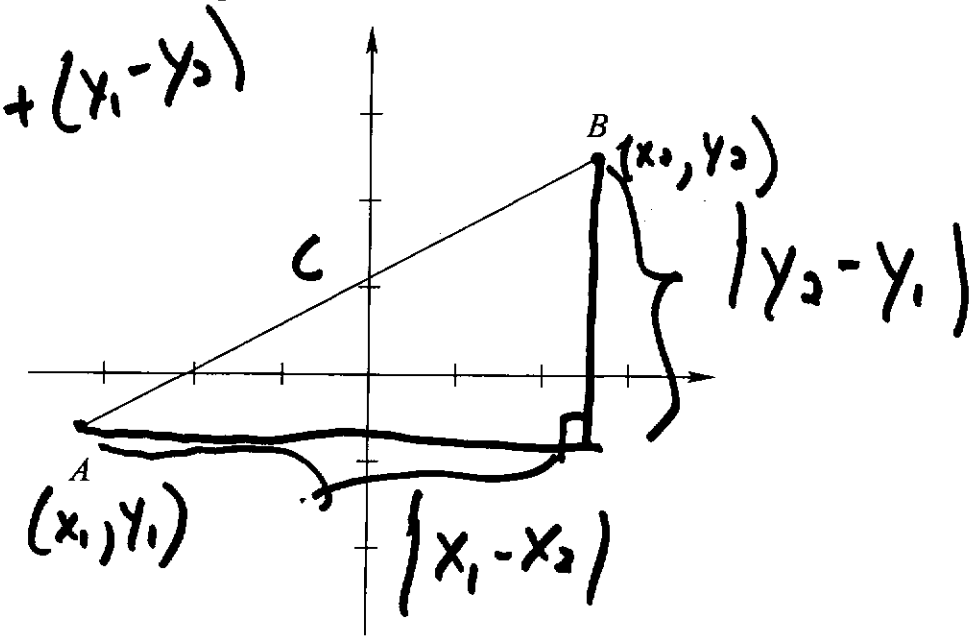
NOTE: The converse of the theorem is also true. That is, if $a^2 + b^2 = c^2$, then the triangle is a right triangle.

Distance Formula

The distance between two points $A = (x_1, y_1)$ and $B = (x_2, y_2)$ is given by

$$d(A, B) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$c^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$



ex. If $P = (-2, 4)$ and $Q = (4, -3)$, find the distance between P and Q.

$$d(P,Q) = \sqrt{(-2-4)^2 + (4-(-3))^2}$$

$$= \sqrt{36 + 49}$$

$$= \sqrt{85}.$$

Midpoints

Midpoint Formula

The midpoint M of the line segment with endpoints $A = (x_1, y_1)$ and $B = (x_2, y_2)$ is

$$(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

ex. Find the midpoint of the line segment that joins $(-1, 2)$ and $(5, 4)$.

$$= \left(\frac{-1+5}{2}, \frac{2+4}{2}\right) = (2, 3)$$

ex. The point $(x, -2)$ is the midpoint of the segment with endpoints $(2, -1)$ and $(3, y)$. Find x and y.

$$x = \frac{2+3}{2} = \frac{5}{2}.$$

$$-2 = \frac{-1+y}{2}$$

$$-4 = -1 + y, \quad y = -3.$$

Lecture 7, Part II: Section 1.2
Graph of Equations

An **equation in two variables** x and y is a statement in which two expressions involving x and y are equal.

An ordered pair (a, b) is a **solution** of an equation in two variables if the equation is true when a is substituted for x and b is substituted for y.

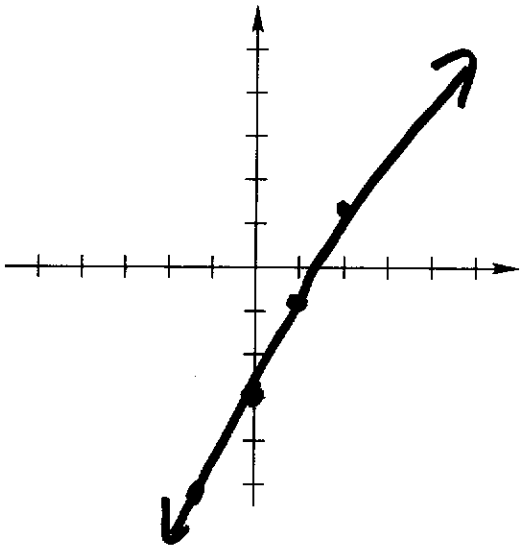
The **graph** of an equation in two variables is the set of points in the xy-plane which are solutions to the equation.

Graph by Plotting Points:

1. If possible, rewrite the equation so that one of the variable is isolated on one side of the equation.
2. Make a table of values showing several solution points.
3. Plot these points on a rectangular coordinate system.
4. Connect the points with a smooth curve or line.

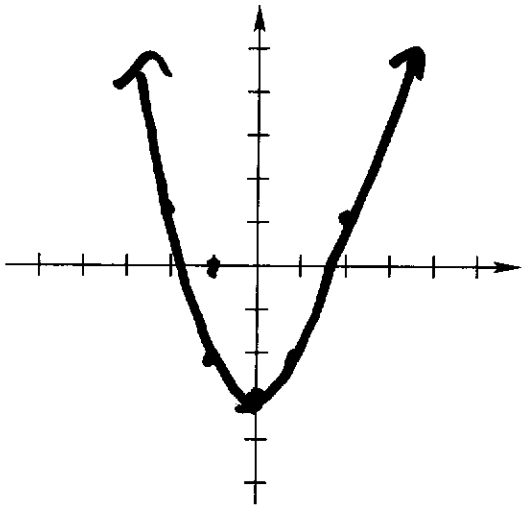
ex. Graph by plotting points:

1) $2x - y = 3$ $\rightarrow$ $\underline{2x - 3 = y}$



x	y
-1	-5
0	-3
1	-1
2	1

2) $y = x^2 - 3$



x	y
0	-3
-1	-2
1	-2
-2	1
2	1

Intercepts

Def. The point(s) at which a graph intersects (crosses or touches) the coordinate axes are called **intercepts**.

An x-**intercept** is the x-coordinate of a point at which the graph intersects the x-axis.
To find it, set $y=0$, solve for x

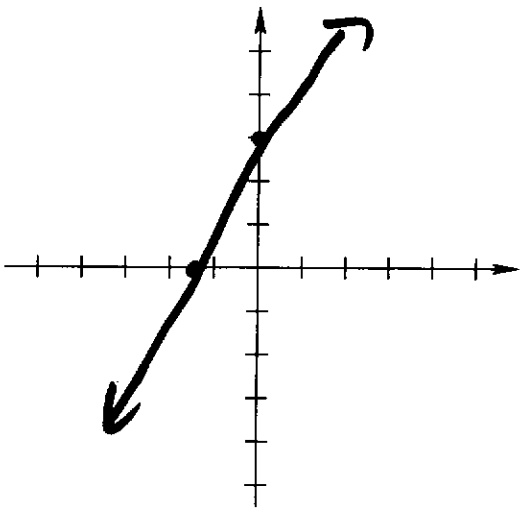
An y-**intercept** is the y-coordinate of a point at which the graph intersects the y-axis.
To find it, set $x=0$, solve for y.

ex. Find the intercepts of the equation and graph:
$y - 3 = 2x \iff y = 2x+3$

x-int: $0 = 2x+3$

$-3 = 2x$

$(-3/2, 0)$ or $-\frac{3}{2} = x$



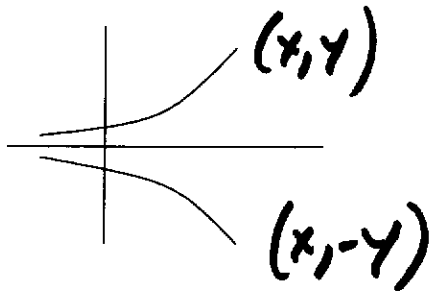
y-int $y = 2\cdot 0 + 3$

$y = 3$

$(0,3)$

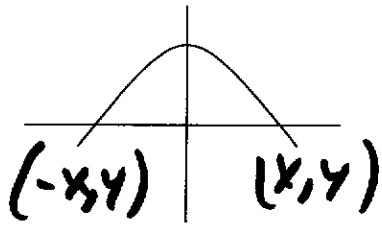
Symmetry

Def. A graph is **symmetric with respect to the x-axis** if for every point (x, y) on the graph, the point $(x, -y)$ is also on the graph.



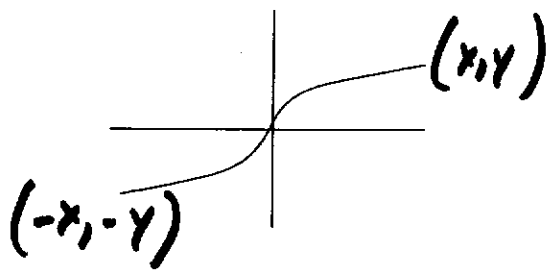
Algebraic test: The graph of an equation is symmetric with respect to the x-axis if replacing y with $-y$ in the equation yields an equivalent equation.

Def. A graph is **symmetric with respect to the y-axis** if for every point (x, y) on the graph, the point $(-x, y)$ is also on the graph.



Algebraic test: The graph of an equation is symmetric with respect to the y-axis if replacing x with $-x$ in the equation yields an equivalent equation.

Def. A graph is **symmetric with respect to the origin** if for every point (x, y) on the graph, the point $(-x, -y)$ is also on the graph.



Algebraic test: The graph of an equation is symmetric with respect to the origin if replacing x with $-x$ and y with $-y$ in the equation yields an equivalent equation.

ex. Test for symmetry and graph the following.

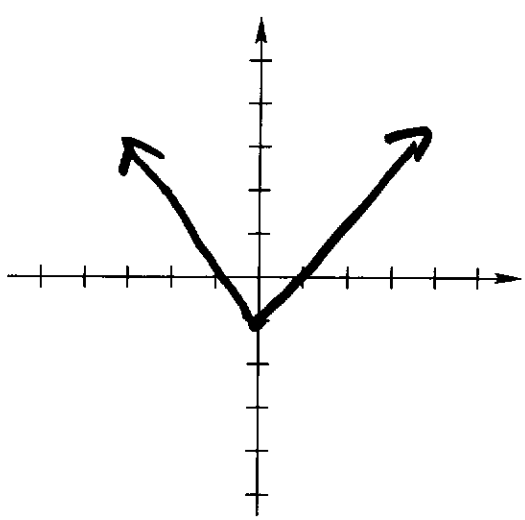
1) $y = |x| - 1$

$x \to -x$

$y = |-x| - 1$

$= |x| - 1$ ✓.

y-axis sym.



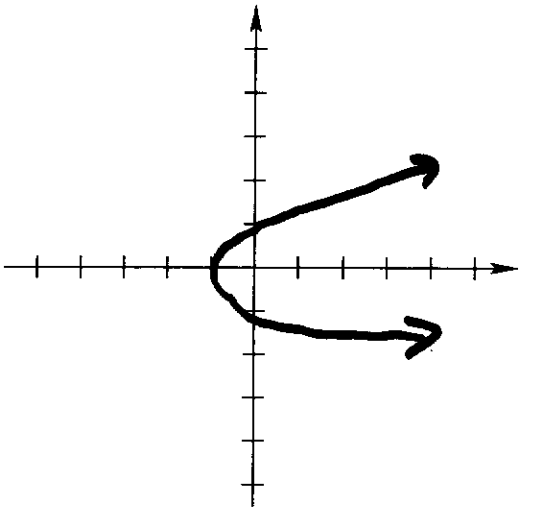
$y \to -y$

2) $x = y^2 - 1$

$x = (-y)^2 - 1$

$= y^2 - 1$ ✓

x-axis sym.

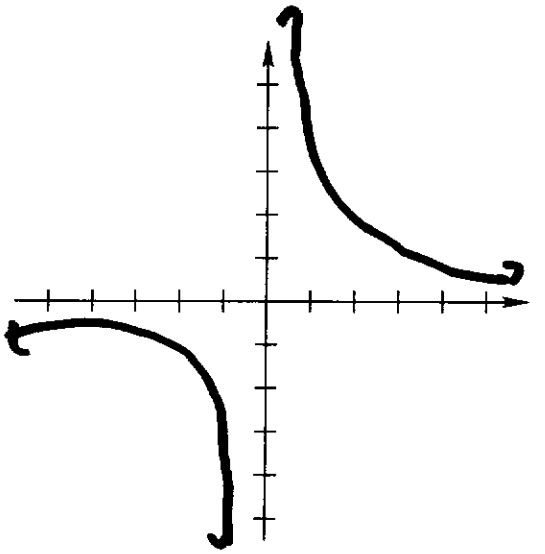


3) $y = \dfrac{1}{x}$

$-y = \dfrac{1}{-y} \Rightarrow$

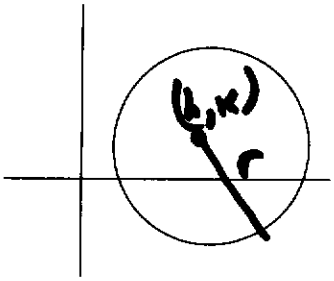
$y = \dfrac{1}{x}$

origin sym.



Circles

Def. A **circle** is a set of points in the xy-plane that are a fixed distance r, called the **radius**, from a fixed point (h, k), the **center** of the circle.



Find a relationship between x and y if $P = (x, y)$ is any point on the circle.

$$d(P, c) = \sqrt{(x-h)^2 + (y-k)^2} = r$$

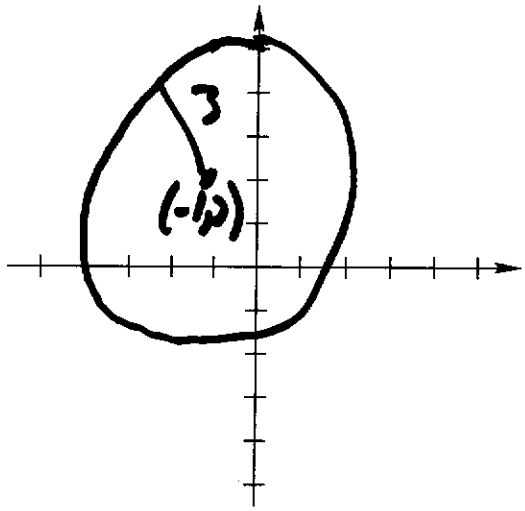
Standard Form of the equation of a circle with radius r and center (h, k):

$$(x-h)^2 + (y-k)^2 = r^2$$

NOTE: The **unit circle** is centered at the origin with radius 1. Its equation: $(h,k) = (0,0)$, $r = 1$

$$\boxed{x^2 + y^2 = 1}$$

ex. Describe the graph of $(x+1)^2 + (y-2)^2 = 9$.

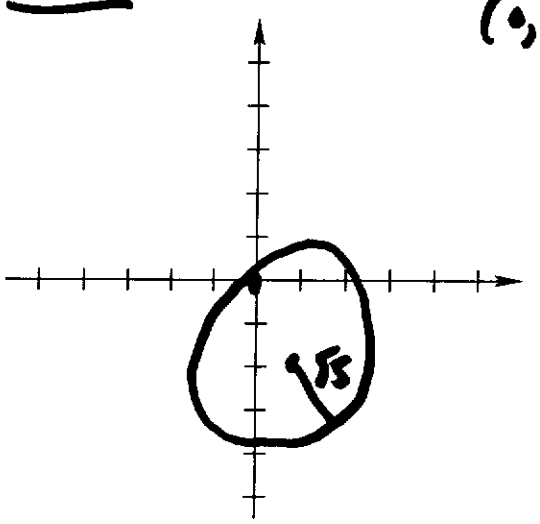


Circle:

$c = (-1, 2)$

$r = 3$

ex. Write the equation of the circle with center $(1, -2)$ and containing the origin. Sketch the circle.



$(0,0)$

$(x-1)^2 + (y+2)^2 = 5$

$c = (1, -2)$

$r = \sqrt{5}$

$d(c, 0) = \sqrt{(1-0)^2 + (-2-0)^2} = \sqrt{5}$

General form of the equation of a circle:

$$x^2 + y^2 + ax + by + c = 0$$

NOTE: If an equation of a circle is in the general form, we use the method of completing the square to put the equation in standard form so that we can identify its center and radius.

ex. Find the center and radius of the circle

$$2x^2 + 2y^2 + 8x - 4y + 9 = 0.$$

$$x^2 + y^2 + 4x - 2y + \frac{9}{2} = 0$$

$$x^2 + 4x + 4 + y^2 - 2y + 1 = -\frac{9}{2} + 4 + 1$$

$$\rightarrow (x+2)^2 + (y-1)^2 = -\frac{9}{2} + 5 = \frac{1}{2}$$

x-

$4 \rightarrow 2 \rightarrow 4$

y-

$-2 \rightarrow -1 \rightarrow 1$

$c: (-2, 1)$

$r = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

Practice.

1) Find all points having an x-coordinate of 2 whose distance from the point $(5, -1)$ is 5.

2) Show that the points $(1, 0)$, $(-1, 1)$ and $(2, 7)$ are the vertices of a right triangle.

3) Find the equation of the circle that has a diameter with endpoints $(-3, 5)$ and $(5, -1)$.

4) Find the equation of the circle center at $(2, -3)$ and tangent to the x-axis (i.e. the circle just touches the x-axis).

Answer. 1) $(2, 3)$, $(2, -5)$ 3) $(x-1)^2 + (y-2)^2 = 25$ 4) $(x-2)^2 + (y+3)^2 = 9$